2.5 Solving Compound Inequalities

Name _____ Period ____ Date _____

Algebra I 2.5 (30-minute plan) · Student sheet

I can…

- write and graph compound inequalities.
- solve compound inequalities that use “and.”
- solve compound inequalities that use “or” and use them in real life.

Vocabulary · fill in the missing word

Compound inequality — two inequalities _____ by “and” or “or”

“And” — the solutions satisfy BOTH _____ (the overlap, often written $a < x < b$)

“Or” — the solutions satisfy AT LEAST ONE _____ (two pieces)

Part 1 · Notes and examples (fill in each note, then work the example)

“And”: “And”: _____ inequalities must be true. Graph the _____.

“Or”: “Or”: at least _____ inequality must be true.

Example 1

Write each as a compound inequality.

a. all real numbers greater than −3 and less than 2

b. all real numbers less than −1 or at least 4

Three-part inequalities: Solve a three-part inequality by doing the same thing to _____ parts.

Example 2

Solve $-4 < x + 3 \leq 9$ and graph.

Example 3

Solve $3x + 1 < -5$ or $2x - 3 \geq 7$ and graph.

Part 2 · Partner practice

Partner practice

Solve $5 \le x - 2 < 10$.